

Semi-Decentralized Multi-Spacecraft Collision Avoidance under Communication Constraints

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Outline

The Problem

LEO is getting crowded

- Thousands of active satellites today
- SpaceX orbital data center: up to 1M satellites proposed
- More conjunctions, more CA decisions needed

Current CA operations

- Ground-driven, operator-specific
- Each operator gets CDMs, plans independently
- Limited by their own ground contact schedule

The Gap

- Fully centralized: assumes perfect info
- Fully decentralized: ignores interaction
- **Neither models intermittent communication directly**

Model spacecraft CA as a
Semi-Decentralized POMDP (SDec-POMDP)
solved with **RS-SDA***

- Ground-station contact windows = synchronization triggers
- Agents share beliefs & coordinate *only* during contacts
- Bridges centralized (always sync) and decentralized (never sync)
- Three scenarios quantify how communication structure shapes safety

State Space

$$s = (d_{\text{miss}}, k)$$

d_{miss} = predicted miss at TCA
(discretized)

k = planning stage (ground contact index)

Bin	Range	Risk
0	[0, 1) km	Collision
1	[1, 5) km	High
2	[5, 20) km	Moderate
3	[20, 100) km	Low
4	[100, 500) km	Nominal
5	[500+) km	Safe

37 states total (6 bins $\times$ 6 stages + sink)

Why miss distance?

- What CDMs report; directly tied to reward
- Collapses 6D relative state to 1 scalar
- No comms variables needed — sync is via contact-window triggers

WAIT $\Rightarrow$ same bin (key invariant)

Under no maneuver, predicted miss at TCA is unchanged — Keplerian dynamics are deterministic. Starting in bin 0 means a collision *will* happen unless a burn is applied before TCA.

Planning Horizon & Rollout Timeline

“Init bin 0 at stage 0”

At $T - 23.4$ h, predicted miss < 1 km.
If nobody burns $\Rightarrow$ collision at TCA.

*When and how should each operator burn,
given only noisy GS updates?*

Contact windows from Brahe

Goddard GS (39°N, 77°W), 10° min elevation,
550 km LEO, 55° inclination.

Stages = midpoints of SC1 passes over Goddard.

Stage	$T - \dots$	Notes
0	23.4 h	First pass; act early
1	9.5 h	14h gap (out of range)
2	7.8 h	Passes cluster here
3	6.2 h	
4	2.8 h	Last meaningful burn
5	1.1 h	Small dv effect
TCA		Reward fires

Why the 14h gap? At 55° inclination over a mid-latitude GS, the spacecraft is out of range for roughly half an orbit period.

Actions & Observations

Actions (per agent)

- WAIT — no maneuver
- $+\Delta v_T$ — prograde burn
- $-\Delta v_T$ — retrograde burn

Joint action space: $3^2 = 9$

Burns are impulsive, along-track, at fixed decision epochs (GS contact midpoints)

Observations (per agent)

Noisy miss distance estimate during ground contact:

$$\tilde{d}_i \sim \mathcal{N}(d_{\text{miss}}, \sigma_{\text{obs}}^2)$$

$\sigma_{\text{obs}} = 5 \text{ km}$ (operational CDM quality)

Between contacts: no update — agent acts on stale belief

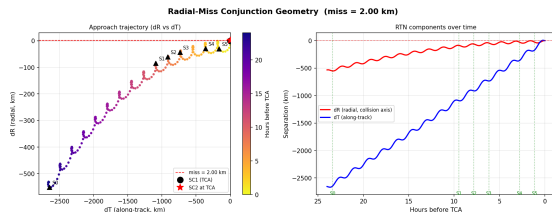
Conjunction Geometry: Radial-Miss

RTN frame centered on SC1:

- **R** (radial, outward): *collision axis* — SC2 offset here
- **T** (along-track): SC2 approaches from behind at $v_{\text{rel}} \approx 15 \text{ m/s}$
- **N** (cross-track): ≈ 0 throughout

At TCA:

$$\delta \mathbf{r}^{\text{TCA}} = \begin{bmatrix} d_{\text{miss}} \\ 0 \\ 0 \end{bmatrix}, \quad \delta \mathbf{v}^{\text{TCA}} = \begin{bmatrix} 0 \\ -v_{\text{rel}} \\ 0 \end{bmatrix}$$



Δv_T (along-track burn) shifts d_{miss} predictably

Along-track burns differentiate bins at all planning stages

Why not crossing geometry?

Crossing = miss in N (cross-track). N collapses to a single bin — collision zone is unreachable from nominal RTN trajectories. Radial-miss puts the collision axis directly in state.

Transition Model

Built offline with Brahe — RS-SDA* never calls Brahe

For each $(d_{\text{miss_bin}}, k, a)$:

- 1 **Place SC2 at TCA** with target miss distance in RTN, then back-propagate to stage epoch τ_k
- 2 Apply joint maneuver (a_1, a_2) impulsively to SC1 and SC2
- 3 **Brahe-propagate** both spacecraft from τ_k to TCA
- 4 Measure resulting miss distance $d' \rightarrow$ bin it $\rightarrow$ next state

Why back-propagation?

Placing SC2 at a radial offset *at the stage epoch* produces the wrong miss at TCA — the offset evolves under Keplerian dynamics, diverging by thousands of km over 24 h. Back-propagating from TCA guarantees the representative trajectory has exactly the target miss distance (< 1 m error across all bins and stages).

Reward Function

$$R(s, a) = R_{\text{collision}}(s) + R_{\text{maneuver}}(a) + R_{\text{comms}}(a)$$

- **Collision penalty** (terminal):

$$R_{\text{collision}} = \begin{cases} -1000 & d_{\text{miss}} < 1 \text{ km} \\ 0 & \text{otherwise} \end{cases}$$

- **Maneuver cost** (fuel / trajectory deviation proxy):

$$R_{\text{maneuver}} = -10 \cdot \mathbf{1}[a_1 \neq \text{WAIT}] - 10 \cdot \mathbf{1}[a_2 \neq \text{WAIT}]$$

- **Communication cost** (Scenario 3):

$$R_{\text{comms}} = -c_{\text{sync}} \cdot \mathbf{1}[\text{sync at this stage}]$$

[TODO: Add trajectory deviation: penalize change in orbital elements accumulated over all maneuvers]

Synchronization Triggers

Ground-station contacts define **when agents can share beliefs and coordinate**

Contact windows computed with Brahe from real GS geometry (Goddard, 550 km LEO, 55° incl.)

Stage	Time before TCA	Scenario 1	Scenario 2
0	$T-23.4$ h	Joint	Joint
1	$T-9.5$ h	Joint	Joint
2	$T-7.8$ h	Joint	Joint
3	$T-6.2$ h	Joint	SC1 only
4	$T-2.8$ h	Joint	SC1 only
5	$T-1.1$ h	Joint	SC1 only

Scenario 1: both SC in similar orbits $\Rightarrow$ all 6 passes are simultaneous $\Rightarrow$ always joint

Scenario 3: all stages available but sync costs c_{sync} per event

Scenario 1: Cooperative Shared Operator

- Both SC controlled by one operator, single ground station (Goddard)
- All 6 contacts are joint sync triggers → always centralized

Experiment

Vary ground network density:

- **Dense** (6–12 stations globally): many syncs, near-centralized
- **Regional** (1–3 stations): fewer syncs, more decentralized

Baselines: centralized (perfect info), decentralized (no sync), SDec (ours)

Expected result: $V_{\text{dec}} \leq V_{\text{sdec}} \leq V_{\text{cen}}$, with $V_{\text{sdec}} \approx V_{\text{cen}}$ using few syncs

Scenario 2: Asymmetric Control

- SC1: fully controllable (our agent)
- SC2: follows a **fixed third-party policy** (unknown to SC1)

SC2 policy variants

- *Uncooperative*: never maneuvers
- *Reactive*: maneuvers if predicted miss $< \theta$ at its contact, θ unknown to SC1

SC1 plans **conservatively** under uncertainty about SC2's behavior and maneuver timing

[TODO: Define how SC2's fixed policy is encoded in transition matrix]

Scenario 3: Costly Communication

- Same contact structure as Scenario 1
- Each sync event incurs penalty c_{sync} in reward

Trade-off

- **Sync** → better joint belief, coordinated action, safer
- **Sync** → costs c_{sync} per event

Experiment: sweep c_{sync} from 0 to 100

Measure: optimal sync count and resulting miss distance

Key question: what is the minimum communication required for safe CA?

Three separate components:

- 1 **Matrix Builder** (offline, once)
spacecraft_matrices.py
Brahe $\rightarrow$ T, O, R $\rightarrow$.npz
- 2 **Policy Solver** (offline)
sdec_spacecraft.py
.npz $\rightarrow$ RS-SDA* $\rightarrow$ policy
- 3 **Simulator** (evaluation)
spacecraft_simulator.py
Policy + Brahe $\rightarrow$ metrics

Key separation

RS-SDA* **never** calls Brahe.

Brahe runs offline to build matrices, then again in the simulator to evaluate the policy with real ECI states.

Simulator metrics

- Minimum miss distance at TCA
- Total Δv accumulated
- Sync event count & timing
- *[TODO: Trajectory deviation]*

Implementation Status

Infrastructure (all done)

- GS contact windows (`test_contacts.py`)
- Conjunction geometry (`test_conjunction_geometry.py`)
- Miss-distance discretizer
- Matrix builder (T, O, R via Brahe)
- Policy solver (RS-SDA* wrapper)
- Simulator (RS-SDA* + greedy rollout)

Experiments & tuning (all done)

- Scenario 1 comparison (100 rollouts, 3 variants)
- Scenario 3 costly-comms sweep ($c \in [0, 200]$)
- Bin-1 terminal penalty ($R = -200$)
- Obs noise tuned to $\sigma = 5$ km
- Policy grid visualization

Pending

- Scenario 3 structural fix (optional sync)
- Scenario 2 (asymmetric control)
- Reward shaping, finer bins

Scenario 1 Results (after bin-1 fix)

100 rollouts, $dv = 0.5 \text{ m/s}$, $\sigma_{\text{obs}} = 5 \text{ km}$, bin-1 penalty = -200

Policy	Init bin	Coll.%	Mean miss	Mean dv	Syncs
Centralized	0	0.0	127.8 km	0.50 m/s	6.0
Decentralized	0	0.0	128.0 km	0.50 m/s	0.0
SDec	0	0.0	127.8 km	0.50 m/s	6.0
Centralized	1	0.0	127.9 km	0.50 m/s	6.0
Decentralized	1	0.0	128.1 km	0.50 m/s	0.0
SDec	1	0.0	127.9 km	0.50 m/s	6.0

Current limitation

Deterministic dynamics + one decisive early burn = all three variants identical. The POMDP is not yet hard enough to differentiate centralized vs. decentralized. Next: stochastic T, finer

Scenario 3: Structural Issue Found

Sync count fixed at 6 regardless of c_{sync}

Policy value degrades linearly ($-10, -130, \dots, -1210$) but sync count stays at 6. Root cause: all states are `sync_states` $\Rightarrow$ sync is mandatory, not optional.

Fix required: model sync as a *choice* during contact windows via `sync_actions` or `sync_observations` in RSSDA, so the policy can trade off communication cost against collision risk.

Next Steps

- 1 **Scenario 3 fix:** make sync optional — encode “communicate” vs. “don’t communicate” as a choice using `sync_actions` in RSSDA API
- 2 **Scenario 2:** encode SC2’s fixed (uncooperative) policy in T matrix:
$$T_{\text{asym}}[a, s, s'] = T_{\text{full}}[a_1 \cdot 3, s, s']$$
- 3 **Reward shaping:** continuous miss-distance penalty $R(d) = -1000 \cdot \exp(-d/d_{\text{safe}})$ instead of step thresholds
- 4 **Trajectory deviation cost:** penalize total Δv / orbital element change
- 5 **Finer bins:** 8 bins — split collision and high-risk zones
- 6 **Obs noise sweep:** compare variants at $\sigma \in \{1, 5, 20\}$ km

Script Run Order

Run all from repo root: `.venv/bin/python spacecraftCA/<script>.py`

Step	Script	Notes
1	<code>spacecraft_matrices.py --force</code>	Build T,O,R (~2 min)
2	<code>spacecraft_discretizer.py</code>	Verify matrices
3	<code>sdec_spacecraft.py</code>	Solve, print value
4	<code>spacecraft_simulator.py --compare</code>	Sc. 1 comparison
5	<code>scenario3_sweep.py --c-max 200</code>	Sc. 3 sweep
6	<code>plot_policy.py</code>	Policy grid (~10 min)
7	<code>plot_conjunction_geometry.py</code>	RTN plot / GIF

Recompile docs from `notes/report/` and `notes/slides/`:
`pdflatex formulation.tex` / `pdflatex slides.tex`

Summary

- Modeled spacecraft CA as SDec-POMDP with ground contacts as sync triggers
- **State**: predicted miss distance at TCA — clean connection to reward, matches CDM reporting
- **Dynamics**: Brahe offline for T matrix; RS-SDA* operates on arrays only
- **Three scenarios** quantify how contact frequency, symmetry, and cost shape safety vs. communication overhead
- **Expected result**: planned synchronizations achieve near-centralized safety with fewer communications